# Metaheuristic Optimization of Mixed-Precision Quantization for Large Language Models

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**Abstract**

Large Language Models (LLMs) are famous for their remarkable abilities to perform well across many natural language processing tasks. However, the high memory usage and computational intensity of LLMs create considerable challenges to production in resource-limited settings. Quantization is one of the most common model compression techniques which reduces the number precision of model parameters and thus reduces memory occupation while possibly accelerating inference. Aggressive quantization can lead to performance degradation, creating the optimization task of choosing proper quantization configurations.

This study aims to explore the application of metaheuristic search algorithms to mixed-precision quantization of Large Language Models , with a focusing on allocating different bit-widths to each model layer. To start with, baseline results are to be obtained using classical uniform quantization methods with simple optimization techniques, for example, random search and greedy search. Next, a few metaheuristic optimization algorithms, such as Genetic Algorithms, Simulated Annealing are to be chosen and evaluated in terms of searching for optimal quantization configurations.

Different quantization approaches will be evaluated based on the memory usage of models, predictive accuracy, inference speed, and optimization cost. Experiments will also explore that if it is possible to use metaheuristics to find quantization.

In the end, we are willing to contribute a experimental evaluation of metaheuristic search strategies for mixed-precision LLM quantization, providing insights into their effectiveness, computational efficiency, and practical applicability.

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# 1 Introduction

LLMs have really made remarkable progress in natural language understanding, text generation, reasoning, and so. However, as these models grow larger, their computational and memory requirements increase as well. This makes deploying them on limited resource devices more difficult. This also increases the overall cost of LLM-based applications.

For reducing the resources required for a model, there are many model compression techniques. One of them is quantization which is used widely by the engineers. This technique represents model parameters with lower numerical precision. For example, instead of storing weights in standard floating-point formats(such as FP32 or FP16), they can be represented using 8-bit or 4-bit values. This can significantly reduce the memory usage of a model while keeping much of its original performance [3, 4].

Post-training quantization (PTQ) can be counted as more useful method because allows us to skip the expensive retraining process. Methods like GPTQ [2] and AWQ [5] have shown that large transformer models can work effectively with low-bit weights. But still, reducing numerical precision has some trade-offs such as approximation errors, which can affect model quality, especially when very low bit-widths are used.

One straightforward approach is uniform quantization, where the same bit-width is applied to all eligible layers of a model. This makes the process a bit simple, but there is a nuance that different layers may react differently to quantization. Some layers can operate with lower precision without much performance loss, while others are more sensitive.

Mixed-precision quantization addresses this problem by assigning different bit-widths to different layers. However, finding an effective configuration is difficult because the number of possible combinations increases exponentially with the number of quantized components. For a model with $L$ configurable layers and $K$ available precision levels, the search space may contain $K^L$ possible configurations.

This creates an opportunity to investigate optimization algorithms that are capable of searching large configuration spaces. Metaheuristic algorithms like Genetic Algorithms (GA), Simulated Annealing (SA) provide search mechanisms that might be suitable for this problem.

The objective of this research is to investigate whether metaheuristic optimization can help to find effective mixed-precision quantization configurations for LLMs. The study will begin with ordinary quantization and simple search strategies. Then it will introduce more advanced metaheuristic algorithms. All experiments will be on the Small Language Models due to computational limitations but may be extended to Large Language Models. The configurations in the result will be compared in terms of model quality, memory usage, inference, and optimization cost.

# 2 Related Work

## 2.1 Neural Network Quantization

Quantization is considered one of the popular methods for improving the efficiency of neural network inference. A broad survey of quantization methods were provided by Gholami et al. [3], covering numerical representations and quantization strategies with their implications for the efficiency and accuracy of the model.

There are common quantization approaches such as post-training quantization (PTQ) and quantization-aware training (QAT). PTQ applies quantization to an already trained model, while QAT adds quantization effects into the training process. Because retraining the LLMs requires considerable computational cost, PTQ is a more practical starting point for researching model compression.

Lang et al. [4] analyzed and discussed the mathematical foundations, implementations, and performance characteristics of the quantization techniques specifically in the context of LLMs. Their work provides a foundation for understanding the limitations of different numerical precisions and motivates researching more flexible quantization configurations.

## 2.2 Post-Training Quantization of LLMs

There are several approaches that are developed to reduce LLM memory requirements and also limit performance degradation.

A post-training weight quantization method called GPTQ(Generative Pretrained Transformer Quantization) was introduced by Frantar et al. [2]. This method uses approximate second-order information to balance quantization errors. GPTQ demonstrates that large transformer models can be quantized to low-bit representations without losing strong predictive performance.

Lin et al. [5] suggested Activation-aware Weight Quantization (AWQ). This method uses activation information to find important weight channels and reduce the effects of quantization. Their findings highlight that model parameters are not equally sensitive to precision reduction.

OmniQuant was introduced by Shao et al. [6] that optimizes quantization-related parameters through learnable weight clipping and equivalent transformations. This work demonstrates that relying exclusively on fixed configurations is not needed. Instead, it highlights the value of optimizing quantization parameters.

Yadav and Bhargavi [7] analyzed post-training quantization for mobile LLM execution and demonstrated the practical benefits of 4-bit quantization for reducing model size. Accordingly this helps with the deployment on resource-constrained devices.

These studies found the practical effectiveness of LLM quantization. The present research will build upon existing quantization methods and, in addition, will examine a separate problem about the selection of effective precision configurations.

### 2.3 Mixed-Precision Quantization and Optimization

The idea behind mixed-precision quantization is assigning different numerical precisions to model components. HAWQ-V2 method proposed by Dong et al. [1], uses second-order sensitivity information to choose configurations. They also added Pareto-based selection to reduce the complexity of assigning bit-widths.

However, sensitivity estimation and configuration selection introduce additional computational costs. Metaheuristic algorithms offer alternative search strategies that do not necessarily require differentiable objective functions and can also explore combinations of distinct bit-width assignments. This research will analyze whether such algorithms can achieve favorable trade-offs between quantization quality and search cost. The comparison will be made against uniform quantization and simpler search-based approaches under controlled experimental conditions.

### 2.4 Research Gap

Existing research shows the effectiveness of post-training quantization and the importance of non-uniform precision allocation. However, to select the right optimization strategy, the size of the configuration space, the cost of evaluation, the model architecture, and the available computational resources must be considered

This study will examine the comparative effectiveness of simple optimization methods and the selected meta-heuristic algorithms for layer-wise mixed-precision quantization of large language models (LLMs). Specifically, the research will focus on whether the improvements in model quality achieved under fixed memory constraints justify the additional computational cost incurred by the optimization process.

## 3 Methodology

### 3.1 Research Design

The research will follow an experimental and comparative methodology. Its implementation will begin with simple quantization configurations. Subsequently, it will introduce optimization-based precision allocation. The study will focus primarily on post-training, weight-only mixed-precision quantization. Activation precision will initially be kept fixed in order to reduce the dimensionality of the optimization problem.

The methodology consists of four stages: baseline quantization, simple search strategies, metaheuristic optimization, and comparative evaluation.

### 3.2 Model Selection and Environment Setup

Experiments will be executed using publicly available small language models most likely from Hugging Face platform. To validate reference measurements

the selected models will first be evaluated in their original precision.

In the experimental environment Python, PyTorch and the required libraries will be used. If possible, experiments will be extended to larger models to investigate the scalability of the proposed optimization approaches.

### 3.3 Baseline Quantization

Initially, experiments will be started with the uniform post-training quantization using selected precision levels. Each quantized model will be evaluated against its original-precision equivalent.This will help us to measure compression and performance degradation, establishing reference results for subsequent mixed-precision experiments.

### 3.4 Simple Search Strategies

Before implementing metaheuristic algorithms, our study will analyze simpler methods for selecting mixed-precision configurations. Random search will generate candidate bit-width assignments. Also greedy search strategy will be used to iteratively adjust layer precision based on estimated performance degradation and memory savings. These approaches will establish comparative foundations and provide insights into the optimization problem.

### 3.5 Metaheuristic Optimization

After the foundational experiments, selected metaheuristic algorithms will be implemented. A Genetic Algorithm (GA) will represent candidate configurations as chromosomes containing precision assignments, using selection, crossover, and mutation operators to refine solutions. Simulated Annealing (SA) will explore the configuration space through iterative modifications and probabilistic acceptance of candidate solutions, not letting it to stuck at local optima.

## 4 Expected Outcomes

The expected outcome of the research is a comparative assessment of simple and metaheuristic search strategies for mixed-precision LLM quantization. The experiments will analyze if metaheuristic methods can be useful to find configurations that keep model quality more effectively than uniform quantization or simple search methods under comparable memory constraints.

## 5 Timeline

This research is planned to be conducted over six weeks, with having four phases.

Table 1: Research Timeline

| **Research Activities** | **W1** | **W2** | **W3** | **W4** | **W5** | **W6** |
|---|---|---|---|---|---|---|
| Model Selection and Environment Setup (3.2) | | | | | | |
| Baseline Quantization (3.3) | | | | | | |
| Simple Search Strategies (3.4) | | | | | | |
| Metaheuristic Optimization (3.5) | | | | | | |

## AI Usage Disclosure

In this research proposal ChatGPT, Claude AI and other generative artificial intelligence tools were used to help with the organization, language refinement, and the latex formatting. The research objectives, methodological decisions, and final content were determined by the author.

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